

# INNOVATIVE APPLICATIONS OF DIGITAL AND INTELLIGENT TECHNOLOGIES IN CLEAN ENERGY

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# CONTENTS



**Background**



**Volatility**



**Inefficiency**



**Systemic  
Challenges**



**Reflections**

# ONE

## BACKGROUND AND CORE CHALLENGES OF CLEAN ENERGY

# THE CLEAN ENERGY PARADOX

## A Counter-intuitive truth

The biggest enemy of clean energy is ?



Capacity can double,  
but if the grid cannot absorb it,  
more capacity becomes stranded  
potential.

# A Counter-Intuitive Truth



Clean energy's biggest obstacle is its own volatility.  
More wind and solar help only if the grid can absorb and use them.



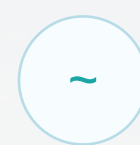
In 2025, nearly 700 GW of new renewable capacity was added globally — yet even in the most advanced economies, the uncertainty of wind and solar output remains the #1 challenge for grid operation.



The question is no longer whether clean power exists; it is whether the variable clean power can be delivered and used when generated.

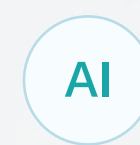


This is not a technology problem — the technologies exist; it is a systems problem, requiring systems-level intelligence.



## Volatility

Intermittent  
wind and  
solar output



## Inefficiency

Hidden losses  
across assets  
and  
operations



## Integration

Distributed  
sources stress  
legacy grids

“ The question is no longer whether clean power exists; it is whether the variable clean power can be delivered and used when generated. ”

## 700 GW

New Renewable Capacity



## #1

Grid Operation Challenge



## Wind + Solar

Volatility Bottleneck



## Systems Intelligence

What Is Needed

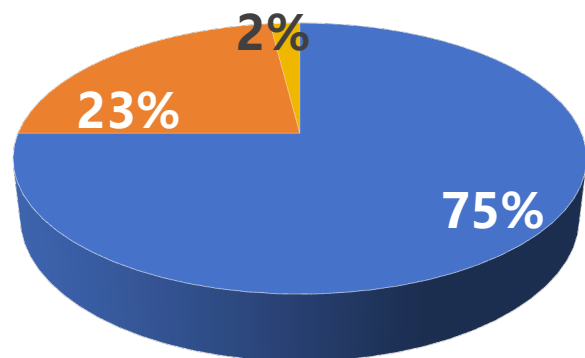


# THE GLOBAL ENERGY TRANSITION AT A CROSSROADS

New renewable capacity added in 2025 (IRENA)

692GW

■ Solar share (511GW) ■ Wind Share (159GW) ■ Others



Every gigawatt of solar makes  
midday oversupply worse

Every gigawatt of wind makes  
the evening ramp steeper

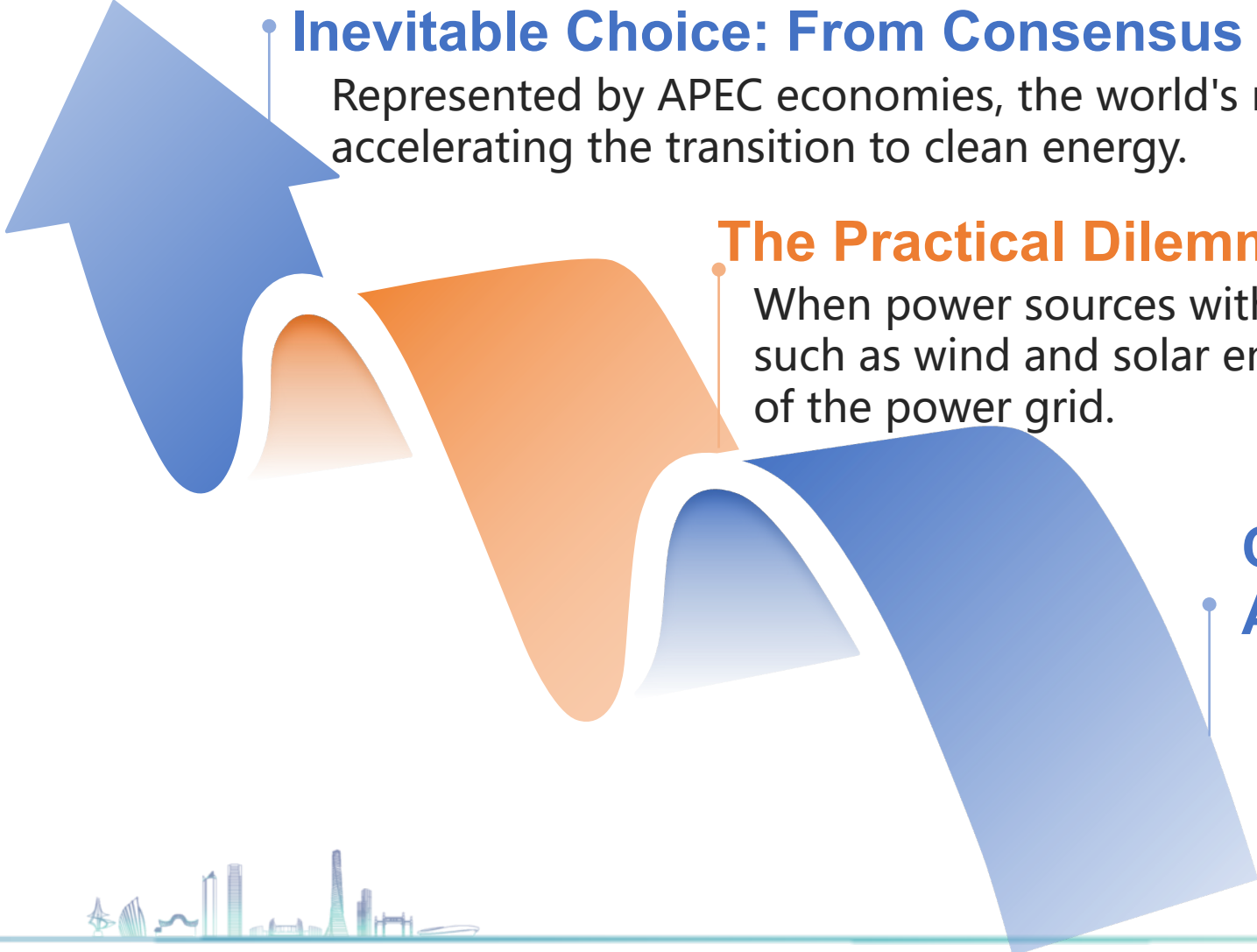
Central  
Paradox



Clean energy is both the solution **AND** the challenge  
—— We are creating the problem even as we solve it



# THE CROSSROADS OF GLOBAL ENERGY TRANSITION



## Inevitable Choice: From Consensus to Action

Represented by APEC economies, the world's major powers are accelerating the transition to clean energy.

## The Practical Dilemma of Intermittent Power Sources

When power sources with strong intermittency and volatility, such as wind and solar energy, gradually become the mainstay of the power grid.

## Core Paradox: The Answer is Also a New Challenge

The large-scale, high-proportion integration also poses unprecedented severe tests to the planning, dispatching and stable operation of the existing energy system.





# Three Core Challenges of Clean Energy

**01 Volatility**

**02 Inefficiency**

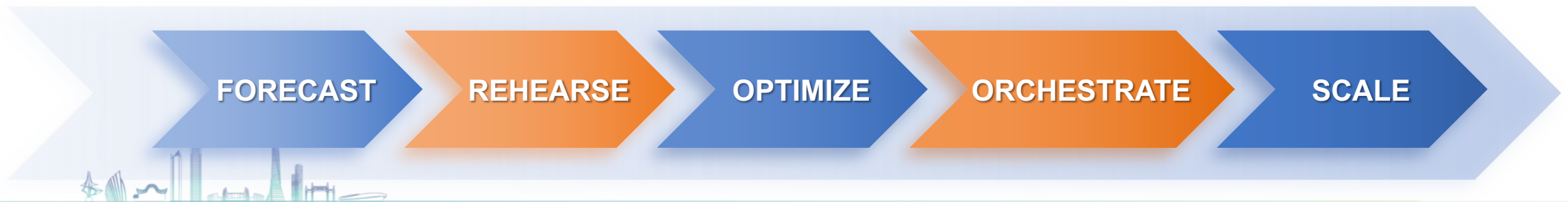
**03 System Integration**



# We need intelligence, not just concrete.



## From Variable Power to Intelligent Systems



# TWO

## ADDRESSING VOLATILITY: AI PREDICTION AND DIGITAL TWINS



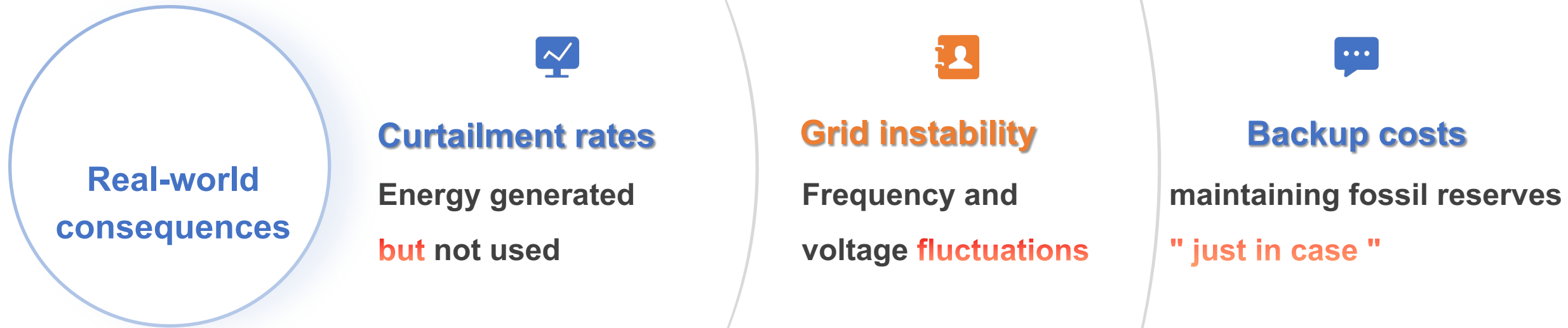
# Challenge 1: Volatility

When the wind stops and the sun sets,  
clean energy needs **foresight** and **rehearsal**.

## Core lens

Systems-level intelligence turns  
**variable** generation  
into **reliable** energy services.

# The fundamental **mismatch** between renewable supply and demand patterns



**Why traditional forecasting falls short:**  
**physics-based models struggle with non-linear weather dynamics**



# How AI Predicts the Unpredictable

## AI forecasting

Reduces forecast errors by 15-25% compared with conventional methods ;  
Can reduce curtailment by over 30% in selected scenarios.

**85%+**

**Wind  
forecast**

State Grid

**91%+**

**Solar  
forecast**

State Grid

**97%+**

**Load  
forecast**

State Grid

**Machine learning  
based forecasting**

**From hours-ahead  
to days-ahead**

**Data Fusion**

**satellite imagery  
+ weather station networks  
+ historical generation data**



**国家电网公司**  
**STATE GRID**  
CORPORATION OF CHINA

# Digital Twins: Simulating Before Operating

AI forecasting tells us what will happen; Digital twins let us rehearse the response.

## Risk Assessment: Preventing Problems



# Key Takeaway from Practice

Forecasting and simulation are not "nice-to-have" — they are the foundation for grid stability at high renewable penetration.

## PRACTICAL INSIGHT

Investment in data infrastructure (weather stations, sensor networks) is as important as investment in clean energy generation capacity.

# THREE

## SOLVING INEFFICIENCY: INTELLIGENT DIAGNOSIS AND OPTIMIZATION

# Inefficiency: Hidden Losses Across the Value Chain

**5-15%**

Generation loss from  
equipment faults

**25-30%**

Offshore wind O&M as  
share of levelized cost

**The cost of "invisible" waste**

dust, wear, and aging  
reduce output over time

suboptimal tracking,  
scheduling, and dispatch

**EQUIPMENT DEGRADATION**

**OPERATIONAL INEFFICIENCY**



# Intelligent Diagnosis: Finding Faults Before They Fail



AI-powered predictive maintenance: from scheduled inspections to **condition-based monitoring**



Computer vision for solar panel inspection  
(drone + AI defect detection)



Acoustic and thermal  
anomaly detection in wind turbines



Industrial-scale deployment across Goldwind's turbine fleet.

- AI predictive maintenance: advance warning **typically days ahead**.
- Maintenance scheduled during low-wind periods, parts ordered ahead of time.
- Turning "reactive repair" into "proactive prevention" — **significantly reduced downtime and maintenance costs**.

THERMAL ANOMALY



ACOUSTIC ANALYSIS



TURBINE FLEET MONITORING

↑ Total Turbines  
1,248

🟢 Online  
1,186

⚠️ Alerts  
28

🔴 Critical  
5

Availability  
97.2%

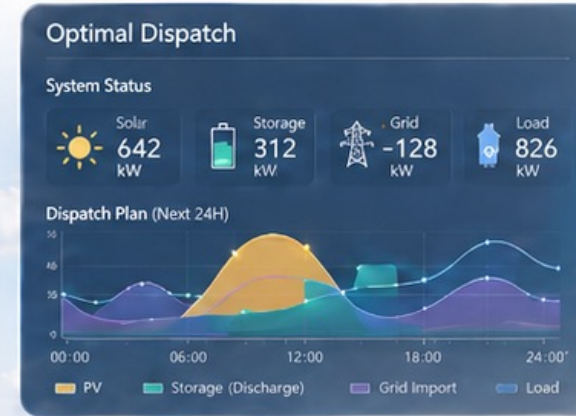


AI DEFECT DETECTION

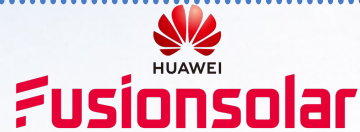
- ✓ Crack
- ✓ Hotspot
- ✓ Delamination

# Optimization: Making Every Kilowatt Count

- AI-based optimal dispatch for hybrid renewable systems;
- Like training a pilot in a **flight simulator** :  
AI tries millions of strategies virtually, learns from failures, applies only the best to the real system;
- **Smart tracking systems for solar PV** :  
AI-adjusted tilt and orientation.



Huawei FusionSolar



**FusionSolar 9.0: edge-cloud full-link intelligence**  
**Significant improvements** in O&M efficiency and storage cost reduction.

**Deep learning** models trained on a vast inverter fleet — fault detection in under **1** second.



# Key Takeaway from Practice

## PRACTICAL FINDING



Efficiency gains from digital intelligence are often **cheaper and faster** than building new capacity.

## TECHNICAL PREREQUISITE



Data transparency and interoperability standards are **necessary conditions** for intelligent optimization.

## RECURRING PATTERN



The technology exists. The question is whether the **data infrastructure and standards** are ready for it.



# FOUR

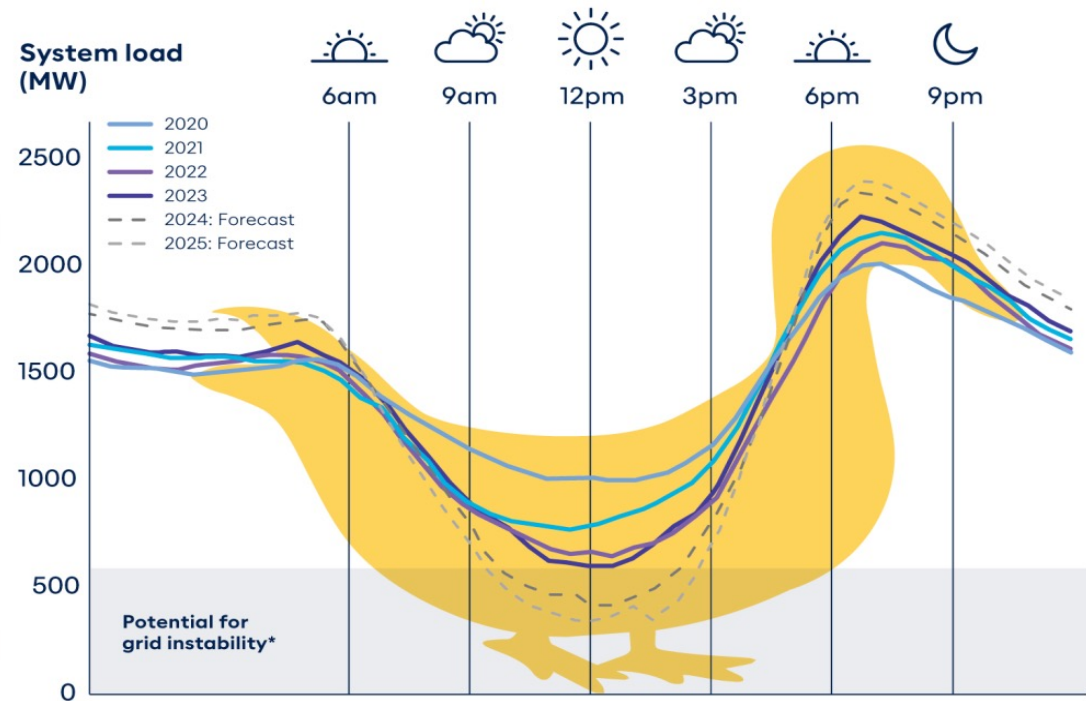
## OVERCOMING SYSTEMIC CHALLENGES: VIRTUAL POWER PLANTS AND SMART MICROGRIDS

# The Problem: The Grid as the Bottleneck

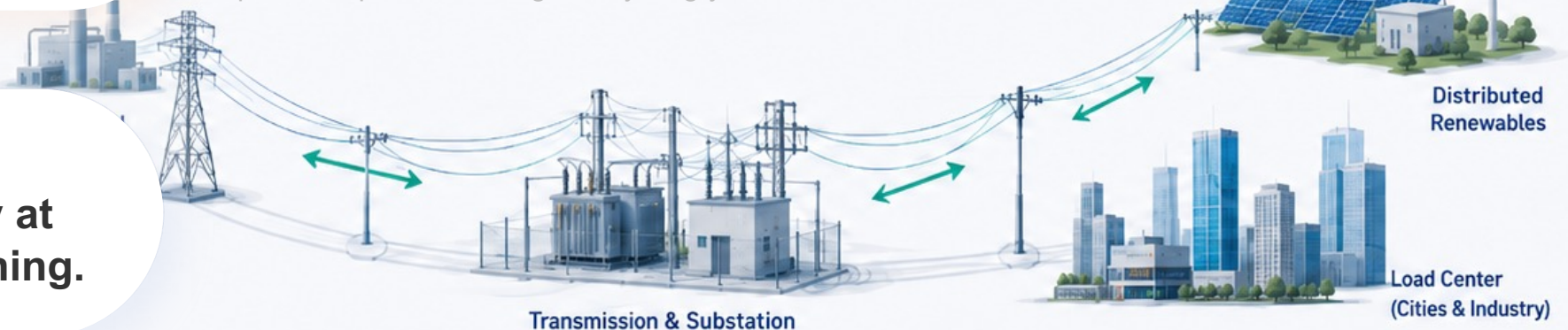
Traditional grids were designed for **centralized, dispatchable** generation.

Distributed renewables create: **bidirectional flows**, **voltage instability**, **frequency challenges**.

The "**duck curve**" and its consequences: oversupply at midday, steep ramp in evening.

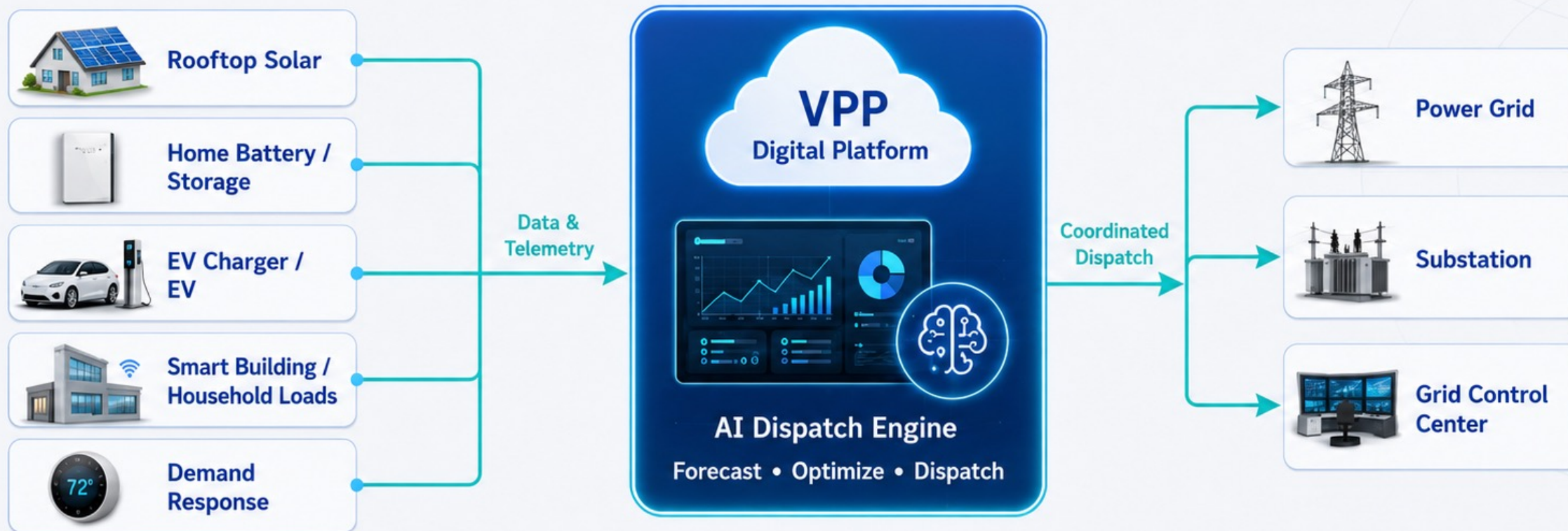


<https://www.pilotech.ai/blogs/everything-you-need-to-know-about-the-duck-curve/>



Duck curve diagram showing net load shape changes with rising solar penetration

# Virtual Power Plants: Orchestrating Distributed Resources



**VPP**

Aggregating distributed energy resources (DERs) through software and AI

**How VPP  
works**

Real-time coordination of solar, storage, EVs, demand response

# Smart Microgrids: Energy Independence at the Local Level

- Microgrid + AI: autonomous operation, self-healing, seamless islanding and reconnection.
- Applications: remote communities, industrial parks, island economies...

## Case: Sanmen Island Zero-Carbon Microgrid (Guangdong, PRC)

Guangdong's **first** "wind-solar-storage" **multi-energy island microgrid**

Serving 800+ residents

Completed in 2025

Ending decades of reliance on **unreliable, expensive diesel-generated electricity**



# Source-Grid-Load-Storage Integration

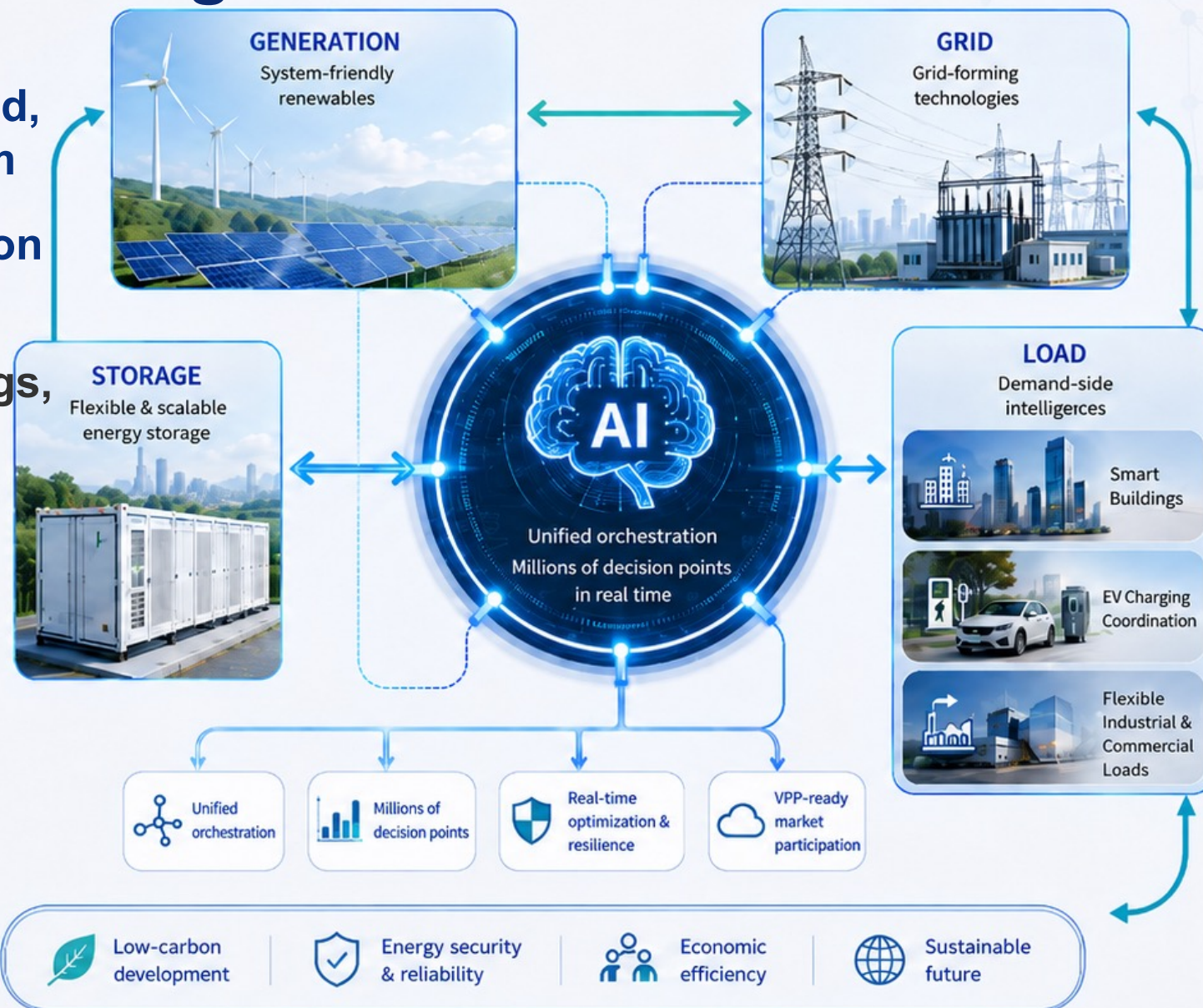
The integrated paradigm: **generation, grid, demand, and storage as a unified system**

AI's role: **coordinating millions of decision points in real time**

Demand-side intelligence: **smart buildings, EV charging coordination, flexible loads**

## State Grid Comprehensive Action Framework

- Grid-forming technologies
- System-friendly renewable stations
- VPPs



# Key Takeaway from Practice

## ⚡ KEY TAKEAWAY

System integration is the **hardest technical challenge** and the **highest-leverage direction**.

## 💡 PRACTICAL INSIGHT

From practice, systemic conditions must keep pace: **data sharing, interconnection standards, and cybersecurity** are prerequisites.

## 🔄 RECURRING PATTERN

Data sharing across organizations, interconnection standards across manufacturers, and cybersecurity across borders.



# FIVE

## PRACTICAL REFLECTIONS AND FUTURE OUTLOOK



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# Mega-City Smart Energy Management Platform



华北电力大学  
NORTH CHINA ELECTRIC POWER UNIVERSITY

## RESOURCE SIDE



HVAC



Data Centers



EV Chargers



Energy Storage



Renewable  
Generation



Power  
Quality



Load



Grid  
Factors



Rich connected  
resources  
Aggregated users:  
4,000+



Mature market  
ecosystem



Response accuracy:  
up to 97%

## MARKET SIDE



Peak Shaving Market



Carbon Market



Electricity Spot Market



Reactive Power Market

Response time  
< 60s



Significant  
economic value



Annual carbon  
reduction:  
510,000  
tons

## Shanghai, China's Largest City-Level Virtual Power Plant

### Real-time tracking

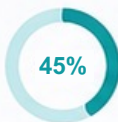


Resource status  
coverage



Forecast  
accuracy

### Operational performance



Load  
Response



Response  
speed



User  
Satisfaction



EV  
Charging



HVAC



Energy  
Storage



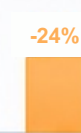
### Value creation



Economy  
(Uni. Revenue)

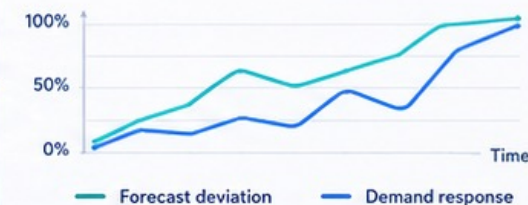


Dispatch  
Efficiency



Emissions  
Reduction

### Accuracy & response



Response speed and precision comparable to real power units



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华北电力大学  
NORTH CHINA ELECTRIC POWER UNIVERSITY

# New-Type Power System



Capability  
Enhancement

Cognitive  
Expansion



光明大模型  
GUANG MING  
LARGE MODEL



Embodied  
Intelligence



World  
Model

## AI-Driven Transformative Innovation



# Our Digital Power System Roadmap



**1,410 GW**

Renewable capacity (end 2024)  
Surpassed coal as #1  
power source



**73.76 GW**

New energy storage  
World's largest  
(40%+ of global)



**42 lines**

UHV transmission lines  
Spanning 50,000+ km



**800B+ RMB**

Grid investment in 2025  
5T+ RMB planned for 15th  
Five-Year

## "New-Type Power System"

— designed for high renewable penetration,  
with digital intelligence as a core dimension,  
not an add-on.



# "AI + Energy" High-Value Scenarios

**51**

High-value scenarios

First batch released May 2026

**8**Major application  
categories**4**Ministries jointly  
issuing action plan**2030**Target year for deeply  
integrated AI-energy ecosystem

## Energy Administration — May 2026

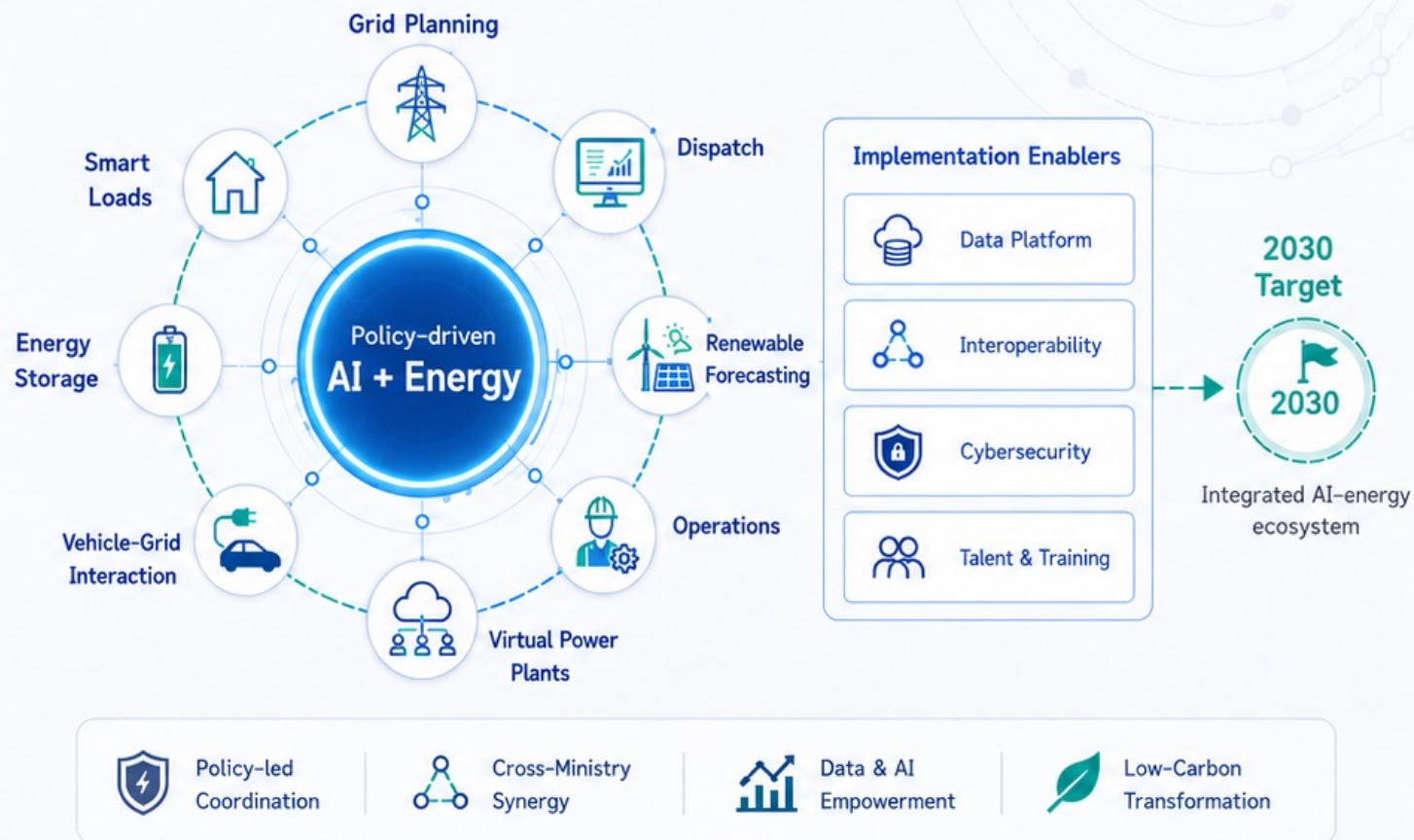
**8 categories include:** grid planning & dispatch, renewable forecasting & operations, virtual power plants, vehicle-grid interaction, and more

Released alongside "AI + Energy" Development Report 2026

Four ministries jointly issued action plan targeting integrated AI-energy ecosystem by **2030**

## STRATEGIC SIGNIFICANCE

We are not treating AI in energy as an experiment. AI is being **systematized** with Energy scenarios, cross-ministry coordination, and the 2030 target.



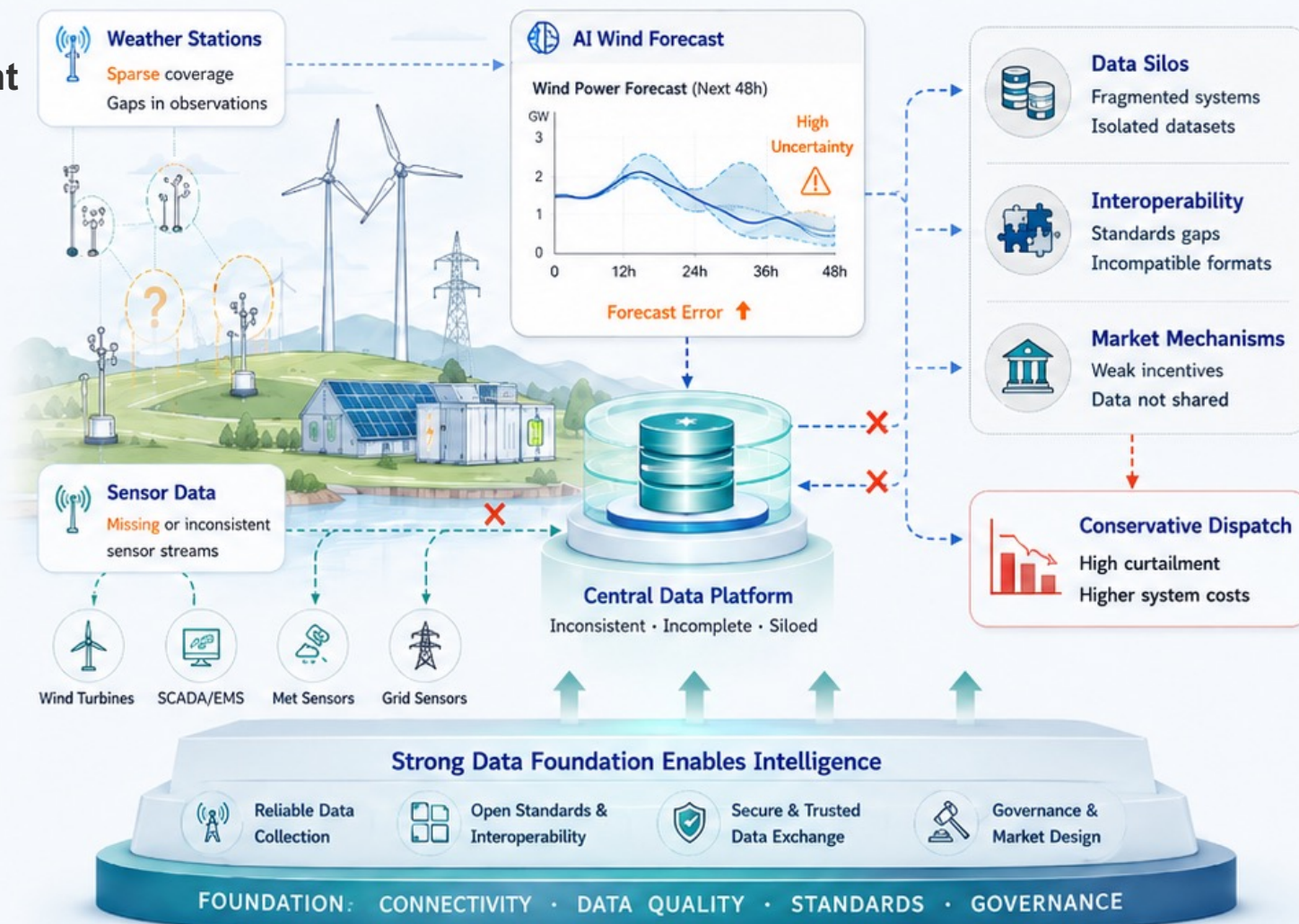
# Data Infrastructure — No Data, No Intelligence

What is still hard — Data silos · Interoperability · Market mechanisms — these are not technology problems, but **systemic conditions**.

- The ceiling of AI forecasting accuracy is often set **not by algorithms but by data** — insufficient weather stations, missing sensor data, and even the best models become powerless.
- **Technical observation:** an APEC economy with insufficient weather station coverage suffered persistently high wind forecast errors, forcing conservative dispatch and high curtailment — not because there was no wind, but because they didn't know when it would come.

## PRACTICAL INSIGHT

Data infrastructure is the **foundation** of digital intelligence — without a solid foundation, even the most advanced technologies cannot be deployed.



# Cross-Border Technical Collaboration

- AI forecasting models need **diverse training data**; APEC economies span scenarios from tropical to arctic, islands to continents — data sharing lets every economy's model "see more weather patterns".
- Cybersecurity** is a prerequisite for cross-border collaboration — digitalization means expanded attack surfaces; standardized security protocols are the foundation for cooperation.

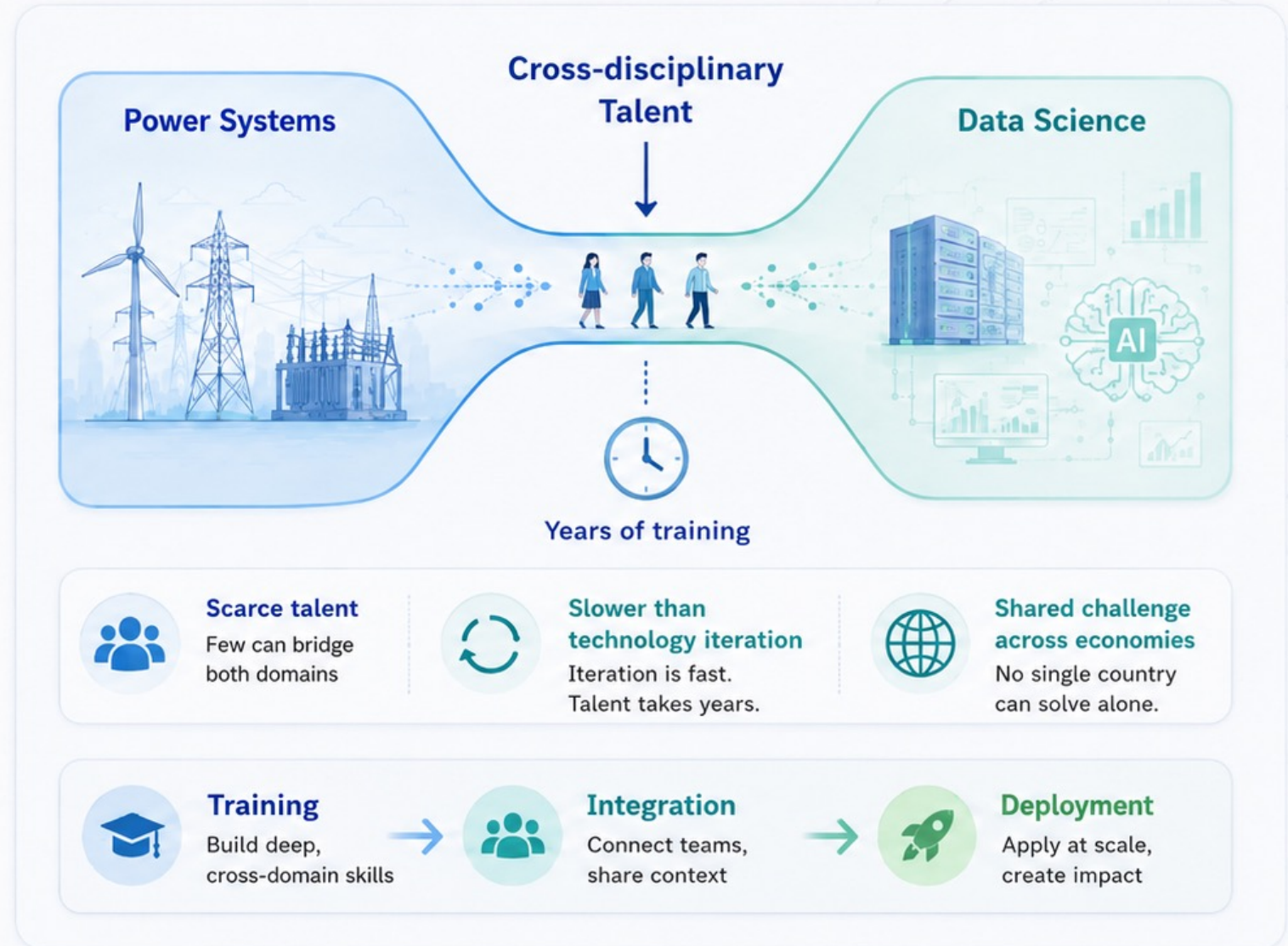
## PRACTICAL INSIGHT

Leveraging **collaboration** platforms to share data, forecasting models and security standards is a practical starting point.



# Talent — A Slower Bottleneck Than Technology

- In our practices, the most deeply felt challenge: talent fluent in **both power systems and data science** is severely scarce — a bottleneck harder to cross than any technology gap.
- Technology can be iterated and upgraded, but **talent development takes time** — this is a shared challenge for all economies.

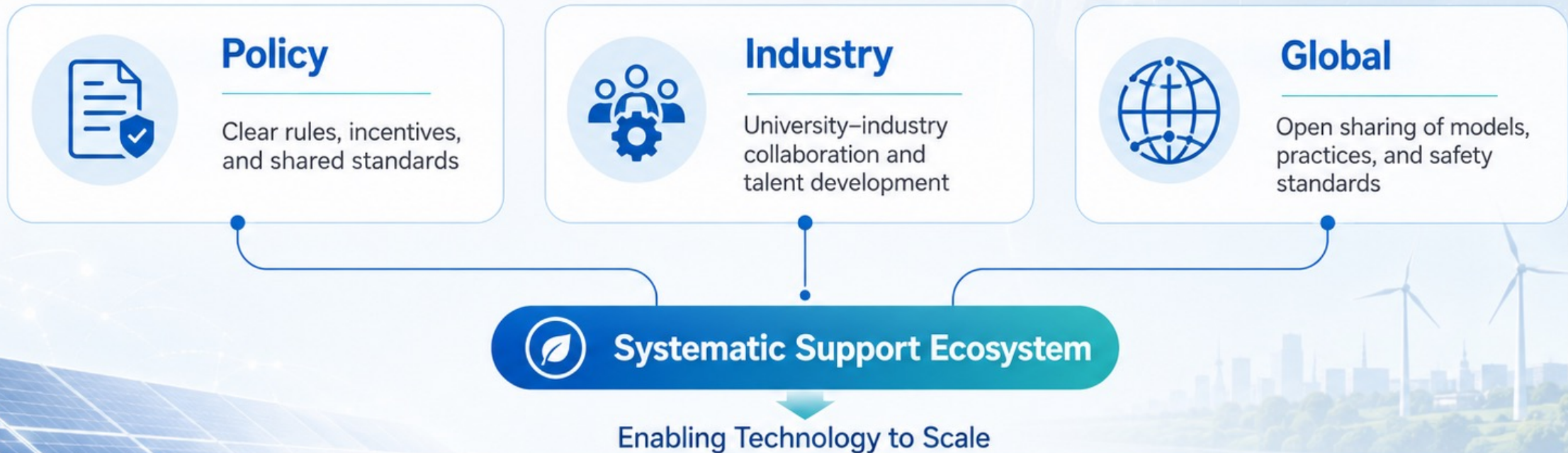


## PRACTICAL INSIGHT

Start cross-disciplinary training **NOW** — this is the one bottleneck where **time is the limiting resource**, not money, not technology.  
**Every year you delay, the gap widens.**

# Future Outlook: Building a Systematic Support Ecosystem

The next step for clean energy is not only better technology, but a stronger ecosystem that helps technology move from lab to large-scale deployment.



“

**Clean energy will scale not only through better technology, but through stronger coordination and ecosystem building.**

”

# THANKS

Welcome your questions and suggestions, let's explore the future of clean energy together.

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